

A Descriptive Statistical Analysis of Audience Reviews for 3 Idiots

Bidisha Garai

June 1, 2026

Presentation Agenda

01. Scope & Data Prep

- Case Study Parameters
- Pipeline & Preprocessing

03. Dispersion & Curves

- Range & Standard Deviation
- Skewness & Frequency Curve

02. Central Tendency

- Mean Sentiment Score
- Median & Mode Metrics

04. Behavioral Dynamics

- Sentiment Category Share
- Review Length Dynamics

Case Study Parameters & Scope

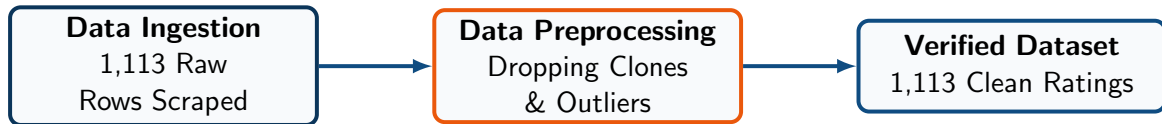
Project Objectives

- Quantify film popularity via descriptive stats
- Apply metrics on custom IMDb metadata
- Clean, parse & analyze textual reviews
- Bridge statistical theory with computation

Selected Film: 3 Idiots (2009)

- **Genre:** Comedy-Drama
- **Runtime:** 2h 50m
- **Theme:** Social pressure in education
- **Why:** Rich audience sentiment patterns

Dataset Collection & Preprocessing



- Source: IMDb (dynamic scroll scraping)
- Removed: duplicate reviews & unrated entries
- **Every rating in this analysis is verified and unique**

8.52

Arithmetic Mean ($\bar{x}$)

Mathematical Formula

$$\bar{x} = \frac{\sum x}{n}$$

- $\sum x$ = sum of all 1,113 ratings
- Scale midpoint = 5.5
- Mean **8.52** is **54.9% above** the midpoint

The average viewer rates 3 Idiots exceptionally high

Median Rating

9.0

- Midpoint of the ordered distribution
- Median (9.0) > Mean (8.52)

Mean is pulled down by a few critical outliers — most viewers actually rated even higher

Mode Rating

10.0

- Most frequently given rating
- **751 out of 1,113** reviews = 10/10

A perfect score is not just common — it is the single most popular opinion

Rating Dispersion & Consensus

Range

$$\text{Range} = X_{\max} - X_{\min} = 10 - 1 = 9$$

- Full scale: 1 to 10
- Critics exist — a **clear minority**

Standard Deviation

$$\sigma = 2.15 \quad (\sigma^2 = 4.62)$$

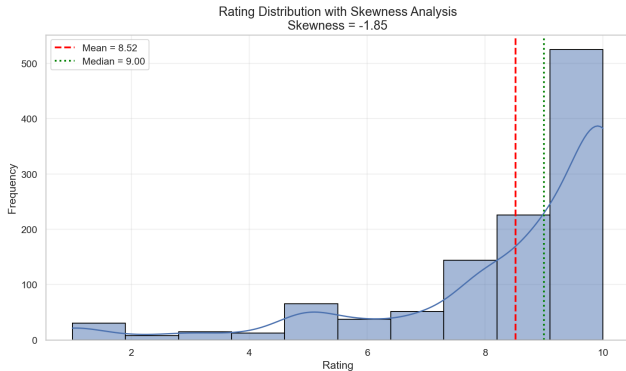
- Low spread around a **high mean**
- Viewers **largely agree** — film is great

Standard Deviation Formula

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$\sigma = 2.15$ confirms high consensus —
not a polarizing film

Skewness Analysis (Distribution Asymmetry)



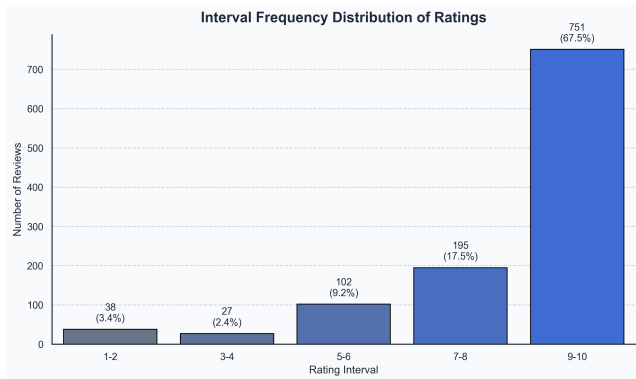
Skewness Formula

$$\text{Skewness} = \frac{\sum (x - \bar{x})^3}{n \cdot \sigma^3}$$

- Skewness = **-1.853**
- **Negatively skewed** — long left tail
- Ratings pile up at **9 and 10**
- Low ratings are **rare exceptions**

**The audience did not just like it
— they loved it**

Interval Frequency Distribution



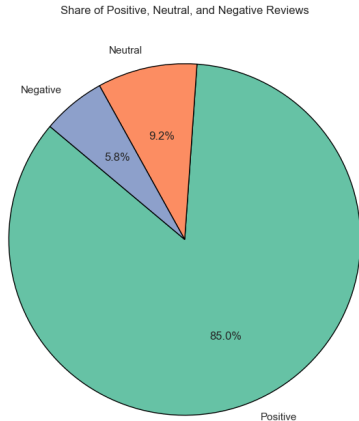
Frequency Breakdown

1-2	38	(3.4%)
3-4	27	(2.4%)
5-6	102	(9.2%)
7-8	195	(17.5%)
9-10	751	(67.5%)

- **85%** of all reviews fall in 7-10
- 9-10 bar **towers over** all others

Audience appreciation is not just positive — it is overwhelming

Sentiment Category Breakdown



Sentiment Quantifiers

Positive: 946 Reviews (85.0%)

Neutral: 102 Reviews (9.2%)

Negative: 65 Reviews (5.8%)

Total N: 1,113

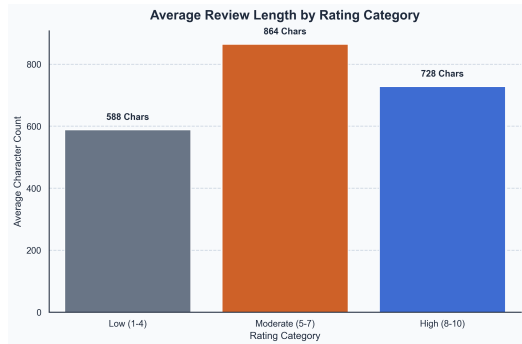
- **85%** positive — strongly dominant
- Negative barely **5.8%**
- No significant polarization

Nearly 9 out of 10 viewers walked away satisfied

Avg. Review Length by Rating Category

- **Low (1–4):** $\approx$ 588 chars
- **Moderate (5–7):** $\approx$ 864 chars (longest)
- **High (8–10):** $\approx$ 728 chars
- Pure praise & pure criticism are **brief**
- Mixed feelings demand **more words**

When viewers are unsure, they write more — certainty needs fewer words



Descriptive Verdict of Final Analysis Questions

Question	Verdict
1. Generally Liked?	Yes — Mean 8.52/10, 85% positive ratings
2. Controversial or Universal?	Universal — $\sigma = 2.15$, strong consensus
3. What does Skewness indicate?	Negative skew (-1.853) — ratings concentrated at high end
4. Max Frequency Interval?	9–10 — 751 reviews (67.5% of dataset)

**” Good analysts calculate data.
Great analysts explain data beautifully.”**

By analyzing 1,113 cleaned IMDb reviews, we move beyond simple ratings to uncover genuine audience behavior. The negative skewness of -1.853, along with the high average rating (8.52) and the most common score (10), proves that the film's success is based on a strong, shared consensus.

Universal Audience Verdict: Highly Appreciated, Highly Successful.